

Solution Requirements

Date	6 Jun 2026
Team ID	XXXXXX
Project Name	Smart Crop
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	The system must securely authenticate Farmers, Researchers, and Policymakers via Multi-Factor Authentication (MFA) or secure OAuth2 protocol providers to protect personalized field data.	High
2	Authorization levels	Define strict RBAC profiles: Farmers: Read/write own soil data, view crop recommendations. Researchers/Policymakers: Access macro regional analytics and trend	High

		pipelines; restricted from editing raw farmer telemetry.	
3	External interfaces	<i>Must connect seamlessly to third-party live meteorology APIs (for temperature, humidity, rainfall metrics) and open agronomic research repositories for matching baseline soil profiles.</i>	Medium
4	Transactions processing	The calculation core must accept real-time transactional input payloads containing Nitrogen (N), Phosphorous (P), Potassium (K), and pH levels, process them through optimization engines, and output structured compatibility rankings.	High
5	Reporting	The platform must generate interactive visual dashboards mapping productivity potential charts, and support exporting data-driven regional strategy summaries as PDF or CSV sheets.	High
6	Business rules, etc.	Optimization recommendations must trigger only when all 7 core environmental fields (N, P, K, temperature, humidity, pH, rainfall) are complete. Crop recommendations must enforce localized soil health preservation boundaries to support sustainable practices.	High

7	Compliance to laws or regulations	All collected geographic and telemetry data must comply strictly with localized data privacy standards (e.g., GDPR frameworks for personal context protection) ensuring user data is anonymized before research aggregation.	Medium
8	<i>Other</i>	Provide automated multilingual localization layers (text and text-to-speech) within the interface so that varied regional farming populations can easily operate the optimization engine.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	The predictive yield and crop recommendation analysis engines must compute results efficiently.	Yield calculation and report generation load within < 2 seconds.
2	Scalability	The system architecture must gracefully scale to handle influxes of active agricultural users during peak seasonal planting windows.	Supports up to 50,000 concurrent active telemetry requests without performance degradation.
3	Security & Data Privacy	All agricultural data payloads, user accounts, and telemetry transmission pipelines must remain encrypted both in transit and at rest.	End-to-end AES-256 encryption; HTTPS protocols across all web endpoints.

4	Reliability & Availability	Provide reliable, round-the-clock availability for field workers and regional planners relying on operational engine guidance.	System achieves a minimum of 99.9% uptime per calendar month.
5	Usability & Accessibility	The digital layout must ensure low-literacy or impaired agricultural stakeholders can navigate the core workflow dashboards easily.	Complies with WCAG 2.1 AA standards; minimum contrast score ratio of 4.5:1.
6	<i>Other</i>	The software code structure must use independent containerized modules to ensure quick updates to individual crop profile configurations.	New crop suitability baseline models can be added or updated with zero core-engine downtime.